Resolution Geometry: Quantum Mechanics

Rigidity, Petz Descent, Task-Declared Geometry, Forced Dynamics Classes, and What Remains Declared

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Abstract

This is the quantum counterpart of the gravitational print. It sets out what RG forces in quantum mechanics, what it types, and what stays declared.

(i) *Rigidity.* Four frozen operational gates (composition descent, extension completeness, in-family closure, continuous extension-stable reversibility) cut an atlas of operational theories strictly down to complex quantum mechanics. The unrestricted uniqueness is imported from reconstruction theorems.

(ii) *Kinematics.* States are preparation quotients. The trace rule is the descent of probability through the effect quotient (Busch–Gleason): ρ is recovered exactly from additive effect data, and a contextual assignment breaks additivity.

(iii) *Petz descent.* A task descends through a channel iff the channel is sufficient. Recovery is canonical, the recovery error is bounded by $2\sqrt{1-e^{-\Delta}}$ in the relative-entropy gap Δ (0/300 violations for the plain Petz map), and the error scales as $\Delta^{1/2}$ (fitted exponent 0.52).

(iv) *Geometry is task-declared.* The monotone metrics form a band, SLD $\leq$ WY $\leq$ BKM $\leq$ RLD. Each is fixed by a task:

- SLD is maximal measured Fisher information (estimation);
- BKM is the Hessian of relative entropy (thermodynamics);
- WY is $8\times$ the skew information (unitary orbits).

The wall behaviour splits on whether $f(0) > 0$.

(v) *Dynamics classes are forced.* The extension gate is complete positivity. Antiunitary symmetries fail it (eigenvalue $-\frac{1}{3}$ at $d = 3$), Stone generators are Hermitian, and the GKLS criterion agrees with complete positivity of the semigroup in 120/120 random generators, in both directions.

(vi) *Hidden versus null in time.* Heisenberg observability separates recoverable hidden components from null ones. Non-Markovian backflow is memory held in an environment of dimension at least the maximal Choi rank.

(vii) *Hamiltonian identification.* One trace with M Bohr frequencies needs $3M+1$ uniform samples, and at $3M$ an alias with different real frequencies matches every sample. One generic probe identifies H locally, while symmetric probes lose exactly their symmetry directions.

What is not derived. The Hamiltonian is selected or measured, as G and Λ are in gravity. Quantum mechanics is conditionally natural with the same theorem shape as GR: the *class* of the law is forced and its *parameters* are declared. An adversarial ledger gives quantum realism its strongest defences (including PBR) and types each count.

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1 Programme and non-circularity

The specification was frozen and hashed before any gate ran (SHA-256 `497fa4b5...`). It contains primitives (preparations, effects, outcome laws, sequential and parallel composition $\boxtimes$, mixing and coarse-graining), the gates, and a candidate atlas. It contains no Hilbert space, Born rule, tensor product, Schrödinger equation or operator algebra. Quantum structure enters only as output. Everything below is either executed (with code) or imported (cited), and the two are kept apart.

2 Rigidity: why complex quantum mechanics

Theorem 2.1 (Rigidity chain; Composition Rigidity)**.** *The candidate atlas contains classical, real, complex and quaternionic quantum theories, boxworld, polygon, random-polygon and superellipse bits. It is cut in two steps:*

- G2 and G3 *(extension completeness and closure) remove real QM (its two-copy kernel is $\sigma_y \otimes \sigma_y$) and quaternionic QM (it has no in-family composite);*
- G4 *(continuous extension-stable reversibility) removes the classical, boxworld and polygon theories, whose reversible groups are finite, and the non-ball smooth bits.*

Complex QM is the unique survivor; none of 300 random polygon theories survives. Tsirelson's $2\sqrt{2}$ appears only as an output.

Theorem 2.2 (Extension–tomography equivalence)**.** *The local quotient of transformations is extension-complete iff composites are spanned by products. In real QM the identity and the transpose agree on every single-system state but differ on a two-system state.*

Imported Theorem 2.3 (Unrestricted reconstruction [1, 2, 3])**.** *Complex QM is the unique finite-dimensional theory satisfying local tomography, continuous reversibility, and the subspace and measurement axioms.*

3 Kinematics: states, probabilities, contexts

Proposition 3.1 (Trace rule as descent; L42)**.** *Suppose a probability assignment is additive on effects, so that it descends through the effect quotient. Then it is* $\mathrm{tr}(\rho E)$ *[4, 5].*

Witness (Q1). The state ρ is recovered exactly from 60 random qutrit effect values. A nonlinear, contextual assignment violates additivity on a POVM split by 0.18. Contextual protocols admit no noncontextual hidden values (L43; Kochen–Specker; 0/512 Peres–Mermin).

4 Information: Petz as RG's descent theorem

Imported Theorem 4.1 (Petz sufficiency [6])**.** *For a channel* $\mathcal{N}$ *and states* ρ, σ*, the following are equivalent:*

- $D(\rho\|\sigma) = D(\mathcal{N}\rho\|\mathcal{N}\sigma)$*;*
- *the Petz map* $\mathcal{P}_{\sigma,\mathcal{N}}$ *recovers both states.*

In RG terms, a task descends through a protocol (a channel) iff data processing is saturated, and the recovery map is canonical.
Witness (Q2). For partial trace of $\rho \otimes \tau$ the gap is 2×10^{-16} and Petz recovery is exact to 5×10^{-16}. Over 300 random channels the gap is always positive.

Theorem 4.2 (Petz modulus)**.**

- Upper bound. *The trace-norm recovery error is at most* $2\sqrt{1 - e^{-\Delta}}$*, where* Δ *is the relative-entropy gap. This is proved for the rotated Petz map [8]. It is executed for the plain Petz map with 0 violations in 300 random channels.*
- Sharp exponent. *Along near-sufficient families the error scales as* $\Delta^{1/2}$*: the fitted exponent is* 0.52 *over gaps from* 3.7×10^{-2} *down to* 10^{-5}*.*

This is RG's identifiability modulus $\omega(\varepsilon)$ *for descent through channels.*

5 Geometry: the quantum metric is task-declared

Imported Theorem 5.1 (Petz classification [7])**.** *The monotone Riemannian metrics on density matrices correspond to operator-monotone functions* f *with* $f(1) = 1$ *and* $f(t) = tf(1/t)$*. They range from SLD (the minimal metric) to RLD (the maximal metric).*

Witness (Q3). In 200 random qutrit states and tangents, SLD $\leq$ WY $\leq$ BKM $\leq$ RLD every time. All four metrics contract under channels, and on commuting data they collapse to classical Fisher information (spread 0).

Theorem 5.2 (Task lock, executed; L39)**.** *Each named metric is fixed by a task:*

(a) Estimation: *the maximum classical Fisher information over measurements equals the SLD metric. It is attained in the SLD eigenbasis, and no random measurement exceeds it (100/100).*

(b) Thermodynamic response: *the Hessian of relative entropy equals the BKM metric. This holds in 100/100 cases, using Richardson extrapolation.*

(c) Unitary orbits: *the WY metric is exactly* $8\times$ *the Wigner–Yanase skew information (constant to* 10^{-8}*).*

Classically, Čencov makes Fisher unique; quantum monotonicity leaves a band. RG's task lock is therefore forced *in QM: declaring the task declares the metric.*

Proposition 5.3 (Walls; L41)**.** *At a rank wall the coherence cost stays finite iff* $f(0) > 0$*. SLD stays at* 4.0 *and WY converges to* 7.98*, while BKM grows logarithmically and RLD as* $1/\varepsilon$*.*

6 Dynamics: the classes are forced

Theorem 6.1 (Extension gate = complete positivity)**.** *At $d = 3$:*

- *unitary conjugations survive extension (minimum eigenvalue -3×10^{-16});*
- *antiunitary ones fail (maximum $-\frac{1}{3}$);*
- *a sampled continuous unitary group has a Hermitian Stone generator (error 10^{-15}).*

In addition, for 120 random Hermiticity-preserving, trace-annihilating generators,

$$P_\perp C_{\mathcal{L}} P_\perp \geq 0 \iff e^{t\mathcal{L}} \text{ is completely positive for all sampled } t,$$

with agreement in all 120 cases (31 conditionally completely positive). The flow $(1-p)\rho + p\rho^{\mathsf{T}}$ is positive but not completely positive. Closed dynamics is therefore unitary with a Hermitian generator, and open Markovian dynamics is of GKLS form [9, 10, 11, 12].

Proposition 6.2 (Hidden versus null in the Heisenberg picture; Q5)**.** *Measuring only $\langle\sigma_z\rangle(t)$ under $H = \sigma_x$, the observable Krylov span is $\{\sigma_z, \sigma_y\}$. The component r_y is a hidden value recovered from the history, while r_x commutes with H and is null.*

Proposition 6.3 (Non-Markovian memory; Q9 and L4)**.** *A qubit exchange-coupled to a qubit environment shows a trace distance that falls to 0 and revives to 0.90. The reduced protocol violates contraction, while the dilated one does not. An exact dilation with a pure environment needs at least $\max_t \operatorname{rank} C(\Lambda_t)$ environment levels (here 2), which the qubit environment attains. General achievability is open.*

Proposition 6.4 (Forgetting; Q7)**.** *Depolarizing contracts by exactly 0.7 per step. Unitary channels do not contract at all, so uniform forgetting requires a declared class (L53–L55).*

7 The Hamiltonian: identifiable, not derivable

Theorem 7.1 (Minimal samples, sharp; L1a)**.** *Consider a real trace $c_0 + \sum_{k=1}^{M}(a_k \cos\omega_k t + b_k \sin\omega_k t)$ sampled uniformly below Nyquist.*

- *The anti-palindromic linear-prediction system has full rank exactly when $N \geq 3M + 1$. The minimum is 4, 7 and 10 for $M = 1, 2, 3$.*
- *At $N = 3M$ a one-parameter family of anti-palindromic characteristic polynomials keeps its roots unimodular. For example, the true frequencies are 0.697 and 0.795, and the alias frequencies 0.730 and 0.788 match every sample to 1.6×10^{-15}.*

Proposition 7.2 (Local identifiability; L1b)**.** *One generic preparation–observable pair has Jacobian rank $d^2 - 1$ for a generic traceless H: 3 for $d = 2$ and 8 for $d = 3$. A symmetric probe (both diagonal) has rank 5 for $d = 3$, losing exactly the directions of its symmetry. Global discrete aliases are not excluded.*

No-Go 7.3 (Hamiltonian selection; the mirror of L128). Every Hermitian H passes every gate. RG forces the *class* (unitary, GKLS) and identifies H with minimal probes, but does not derive it. This is the theorem shape of the gravitational print: the Einstein class is forced, while G and Λ are measured.

8 Main theorem

Main Theorem (Conditional naturality of quantum mechanics). *Assume the frozen gates G1–G4 together with the imported subspace and measurement axioms.*

(i) Complex finite-dimensional QM is the unique operational theory: on the atlas this is executed, and in general it is imported.

(ii) Probability is the trace rule, by descent through the effect quotient.

(iii) Information processing obeys Petz descent, with modulus $\Theta(\Delta^{1/2})$.

(iv) State geometry is a band of monotone metrics, and each task declares one.

(v) Dynamics is unitary (closed) or GKLS (open Markovian). Non-Markovian behaviour is environment memory.

(vi) The Hamiltonian is identifiable with $3M+1$ *samples per trace, and locally from one generic probe, but it is not derivable.*

9 Adversarial ledger for quantum realism

Count	Evidence	Strongest defence	Verdict
The state is a preparation quotient, not a thing	L0–L5; descent of composition (C1)	PBR [13]: under preparation independence the state is ontic	CONTESTED
The Born rule is a consequence, not a law	Gleason/Busch descent (Q1)	Noncontextuality is itself a physical postulate	CONCEDED (conditional)
The dynamics classes are forced	Extension gate (L5, GKLS 120/120)	—	ACQUITTED (forced, not chosen)
The Hamiltonian is selected	L128 mirror	Every theory has parameters	CONCEDED
There is no unique quantum geometry	Petz band; task lock executed	Fubini–Study is the pure-state limit	CONCEDED (mixed states)
Interpretations are one fibre	Equal observation laws (Cornering F12)	PBR and ontological models	CONTESTED
Local hidden values	CHSH ≤ 2 against $2\sqrt{2}$; KS 0/512	Nonlocal or superdeterministic models	CONCEDED (local noncontextual values)
Non-Markovian "memory"	Backflow equals hidden environment (Q9, L4)	—	typed

How this can be wrong.

- A theory outside the frozen atlas that satisfies G1–G4 and differs from QM would break the executed rigidity; the imported theorems exclude it under their axioms.

- A failure of complete positivity in nature, such as initially correlated system and environment, would change the dynamics class.
- A proof that the plain Petz map violates the modulus would weaken that line.

10 The two prints side by side

	Gravity	Quantum
Forced class	linearized and cubic Einstein (declared class)	complex QM; unitary or GKLS dynamics
Gate doing the work	descent and conservation	extension (complete positivity)
Declared or measured	G, Λ, $D = 4$, derivative order	H, subspace and measurement axioms, environment
Geometry	spacetime is a representative	metric is a task-declared band
Hidden versus null	TT modes versus gauge	Krylov span versus commutant

11 Reproduction and limits

Batteries. All reruns are byte-identical.

- `rg_rigidity.py`: the frozen C0 specification is verified first, then C1–C6 run.
- `rg_qm_probe.py`: Q1–Q9.
- `rg_qm_locks.py`: L1–L5.

Limits.

- Finite dimensions only.
- The reconstruction, Petz, Wigner, Stone and GKS–Lindblad theorems are imported.
- The plain-Petz modulus is executed, not proved.
- The identifiability of H is local.
- No experimental data are used.

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